1. Identifying Key Variables and Mathematical Aspects of a Real-World Problem

(Corresponds to Points 2 & 6 in formulate.pdf)

- Question Content: The problem describes a revolving door with an inner diameter of 2 meters, three wings dividing the space into three equal sectors, each holding a maximum of two people. The door rotates four complete times per minute, and the question asks for the maximum number of people entering in 30 minutes.
- Relevance Analysis:
  - Students must identify critical variables:
    - Rotations per minute (4)
    - Number of sectors (3)
    - Capacity per sector (2 people)
    - Time duration (30 minutes)
  - They must also recognize that only people entering through the "Entrance" count toward the total, but due to the symmetric, continuous rotation, each full rotation allows 3 sectors × 2 people = 6 people to enter.
  - This directly aligns with formulate.pdf's emphasis on "identifying key variables" and "describing the mathematical aspects of a real-world problem situation."

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2. Simplifying or Decomposing a Situation for Mathematical Analysis

(Corresponds to Point 7 in formulate.pdf)

- Question Content: A real revolving door involves complex dynamics like human behavior, timing of entry/exit, and potential congestion.
- Relevance Analysis:
  - The problem abstracts away all real-world complexities and assumes ideal conditions:
    - The door is always at maximum capacity (2 people per sector).
    - Every rotation efficiently delivers 6 new people into the building.
    - No time is lost between rotations.
  - This simplification makes the scenario tractable for mathematical calculation, exemplifying formulate.pdf's principle of "simplifying or decomposing a situation to make it amenable to mathematical analysis."

3. Translating a Real-World Problem into a Standard Mathematical Representation or Algorithm

(Corresponds to Points 9 & 10 in formulate.pdf)

- Question Content: The student must convert the physical description into a multi-step arithmetic calculation.
- Relevance Analysis:
  - The translation process is as follows:
    1. People per rotation = 3 sectors × 2 people = 6
    2. People per minute = 4 rotations × 6 = 24
    3. People in 30 minutes = 24 × 30 = 720
  - This sequence represents a standard algorithmic approach to a rate-time-total problem.
  - It demonstrates "translating a problem into a standard mathematical representation or algorithm" and "using mathematical tools to describe relationships."

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4. Recognizing Mathematical Structure and Correspondence to Known Concepts

(Corresponds to Points 5 & 8 in formulate.pdf)

- Question Content: The setup involves repeated cycles (rotations) and fixed capacity per cycle.
- Relevance Analysis:
  - Students should recognize this as a classic rate problem (people per unit time), a well-known mathematical structure.
  - The regularity of the system (constant rotation speed, fixed capacity) allows the use of multiplication instead of more complex modeling.
  - This reflects formulate.pdf's focus on "recognizing mathematical structure (regularities, patterns)" and "connecting the problem to known mathematical concepts or procedures."

1. 识别现实问题中的关键变量及其数学要素

（对应《formulate.pdf》第2条和第6条）

- 题目内容：题目描述了一扇内径为2米的旋转门，由三片翼板将空间均分为三个扇区，每个扇区最多容纳两人。门每分钟完整旋转4圈，问题要求计算在30分钟内最多可进入多少人。
- 相关性分析：
  - 学生必须识别出关键变量：
    - 每分钟旋转圈数（4圈）
    - 扇区数量（3个）
    - 每个扇区容量（2人）
    - 总时间（30分钟）
  - 同时需理解：虽然图中标注了“入口”（Entrance），但由于旋转门对称且连续运转，每完成一圈旋转，就有3个扇区 × 2人 = 6人进入建筑物。
  - 这直接体现了《formulate.pdf》所强调的“识别关键变量”和“描述现实问题情境中的数学要素”。

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2. 简化或分解情境以便进行数学分析

（对应《formulate.pdf》第7条）

- 题目内容：现实中旋转门涉及人流行为、进出时机、拥堵等复杂因素。
- 相关性分析：
  - 题目忽略所有现实复杂性，做出理想化假设：
    - 每个扇区始终满载（2人）；
    - 每次旋转都能高效送入6名新人员；
    - 旋转之间无时间损耗。
  - 这种简化使问题变得可通过基础数学解决，完美体现了《formulate.pdf》中“简化或分解情境以使其适合数学分析”的原则。

3. 将现实问题转化为标准的数学表达或算法

（对应《formulate.pdf》第9条和第10条）

- 题目内容：学生需将物理描述转化为多步算术计算。
- 相关性分析：
  - 转化过程如下：
    1. 每圈进入人数 = 3扇区 × 2人 = 6人
    2. 每分钟进入人数 = 4圈 × 6人 = 24人
    3. 30分钟总人数 = 24 × 30 = 720人
  - 这一序列构成了一个标准的算法化求解流程，属于典型的“总量 = 速率 × 时间”问题。
  - 它体现了“将问题翻译为标准数学表示或算法”以及“运用数学工具描述变量间关系”的能力。

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4. 识别问题中的数学结构并关联已知概念

（对应《formulate.pdf》第5条和第8条）

- 题目内容：情境包含重复循环（旋转）和固定容量。
- 相关性分析：
  - 学生应识别出这是一个经典的速率问题（单位时间内的数量），属于常见的数学模型。
  - 系统的规律性（恒定转速、固定容量）允许直接使用乘法，而无需更复杂的建模。
  - 这反映了《formulate.pdf》所强调的“识别数学结构（如规律、模式）”和“将问题与已知数学概念或方法建立联系”。